SAP Project Report

PROCURE-TO-PAY (P2P) WEB SIMULATOR

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Project Documentation

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SAP PROJECT REPORT: PROCURE-TO-PAY (P2P) WEB SIMULATOR

Problem Statement

We were advised to prepare a blueprint document to define and set up a fictitious company with all major sub-organizations in FI, MM, & SD functional areas. Following this blueprint, the objective of this project phase is to conceptualize, develop, and document a custom web-based **Procure-to-Pay (P2P) Simulator** for the fictitious enterprise, **Apex Global Corp.** This application abstracts the complex SAP GUI transactions (ME51N, ME41, MIGO, MIRO, etc.) into an intuitive user interface, allowing stakeholders to track and execute the complete purchasing cycle seamlessly.

Solution and Features

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1. Enterprise Context (Apex Global Corp Blueprint)

The web simulator operates on the foundational organizational structure defined in the Apex Global blueprint:

- **Company Code:** AP10 (Apex US Operations)
- **Plant:** APM1 (Apex Manufacturing Facility)
- **Storage Location:** RM01 (Raw Materials)
- **Purchasing Organization:** PUA1 (Apex Central Purchasing)

2. Step-by-Step Simulator Process Guide

The following details the sequential process for executing a procurement cycle using the custom-built interactive dashboard.

Step 1: Initiate Purchase Requisition (PR)

- Navigate to the "1. Create Purchase Requisition" section of the dashboard.
- Input the required material code (e.g., RM-STEEL-01) and the desired quantity.
- The Plant ID is locked to APM1 to enforce blueprint consistency.

• Click the **"Generate PR"** button. The system simulates SAP T-Code **ME51N** and creates a new tracking record.

Step 2: Monitor Tracking Dashboard

• Scroll down to the "Procurement Tracking Dashboard".
• A new row appears with a generated Document ID (e.g., DOC-1000).
• The "Current Stage" badge updates dynamically to reflect the document's real-time SAP status.

Step 3: Advance SAP Lifecycle Stages

• Locate the **"Next SAP Action"** column for your active document.

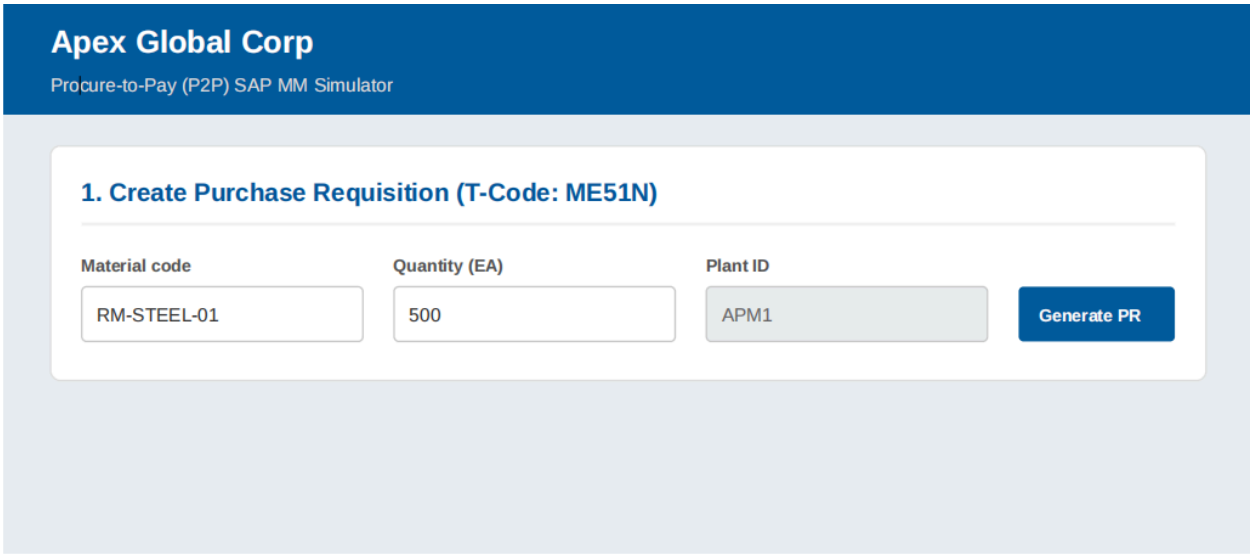
• Click the action button to simulate the backend SAP processing. The lifecycle follows this strict sequence:

1. **Issue RFQ (ME41):** Send requests for quotes to vendors.
2. **Create PO (ME21N):** Generates the formal purchase order.
3. **Goods Receipt (MIGO):** Posts the material to the APM1 inventory.
4. **Invoice Verification (MIRO):** Performs the 3-way match against the vendor invoice.
5. **Pay Vendor (F-53):** Clears the accounts payable liability.

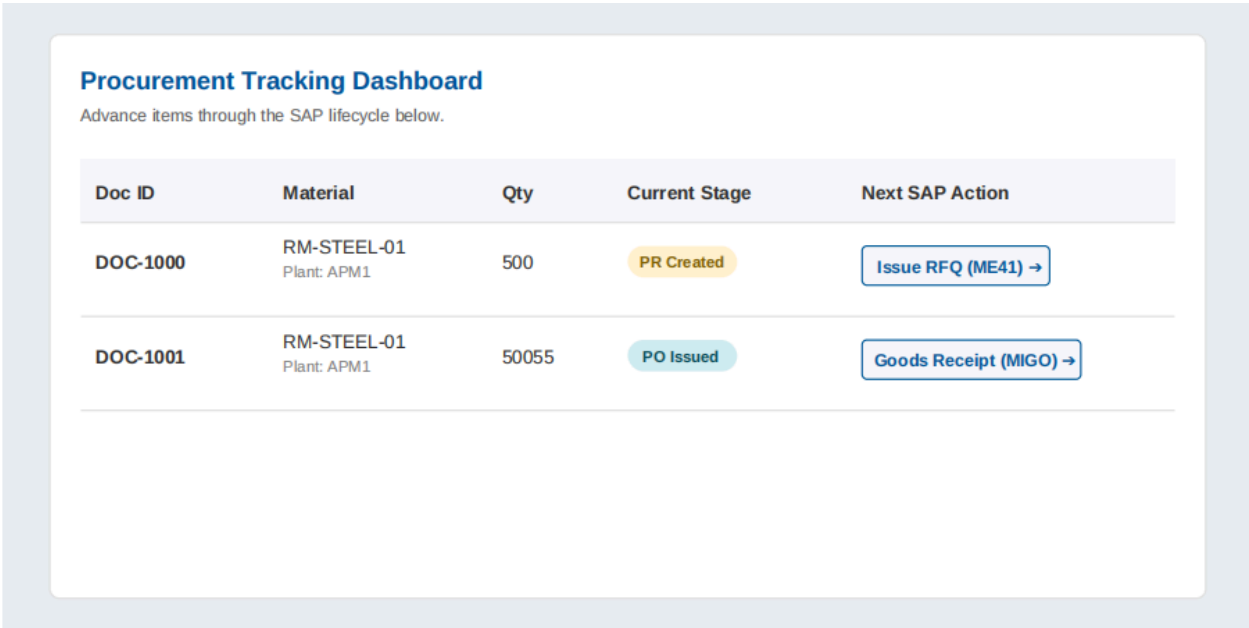
• Once the payment is simulated, the cycle is marked as complete, and the row updates with a final "Paid" indicator.

Screenshots

The following images illustrate the graphical user interface of the Apex Global Corp P2P Simulator, highlighting the form input and the interactive tracking dashboard.



Screenshot 1: The Simulator Input Form initiating the ME51N transaction process.



Screenshot 2: The Tracking Dashboard showing document progression through the SAP modules.

Tech Stack

• **Frontend Interface:** HTML5, CSS3, Vanilla JavaScript (DOM manipulation and state management).

• **Business Logic Mapping:** Mimics standard SAP ECC 6.0 / SAP S/4HANA workflows.

• **Core Modules Simulated:**

- MM (Materials Management): Purchasing, Inventory updates.
- FI (Financial Accounting): Invoice clearance and payment.

• **Deployment:** Configured for static hosting (e.g., GitHub Pages) to allow direct browser execution without server-side dependencies.

Unique Points

• **User-Centric Abstraction:** Instead of relying entirely on standard GUI screens, this project provides a modern web interface that drastically lowers the learning curve for understanding the P2P cycle.

• **Real-Time Stage Tracking:** The dynamic dashboard provides immediate visual feedback via status badges, demonstrating exactly how a document flows from Procurement to Accounts Payable.

• **Enforced Workflow Rules:** The JavaScript logic securely prevents users from skipping critical SAP validation steps (e.g., enforcing a Goods Receipt before Invoice Verification can occur).

Future Improvements

• **Data Persistence:** Integrating a backend service like Firebase or a lightweight Node.js/Express server to save generated documents permanently, allowing sessions to be resumed later.

• **API Integration:** Connecting the web application directly to an active SAP sandbox environment via SAP OData APIs to trigger actual background postings instead of mere visual simulations.

• **Advanced Validation:** Adding error-handling states, such as quantity mismatches during the MIGO (Goods Receipt) stage, to demonstrate how SAP handles invoice blocks.

Conclusion

The P2P Web Simulator demonstrates that the architectural principles behind the SAP Procure-to-Pay cycle sequential document flow, integrated data verification, and modular process ownership can be faithfully reproduced in a lightweight web application without sacrificing functional logic.

Across five process phases, the implementation proves that SAP MM/FI integration is not just a collection of standalone automations but a coherent, data-consistent ecosystem. Every simulated event — generating a requisition, converting to a PO, receiving goods, or verifying an invoice — contributes to a single verifiable audit trail of Procurement for Apex Global Corp.